ECON 0100 | Part E

Vignette E4

Unicorn Horn

Unicorn horn is an important ingredient in many potions voluntarily donated by unicorns at the end of every season. The Apothecary is the only seller with access to unicorns so everyone who needs the substance must buy it at the Apothecary, with demand given by the following equation:

$$D : P = 50 - Q \tag{1}$$

with costs in galleons and quantity in killograms. The marginal cost of producing unicorn horn is a constant 5 galleons per stone and no fixed costs.

Q1 | Profit Maximizing Quantity

Suppose the Appothecary sells unicorn horn to anyone for the same price. Knowing that the marginal revenue generated by each unit sold is $MR = 50 - 2Q$, calculate the Apothecary's profit maximizing quantity.

Q2 | Profit Maximizing Price

What is the Apothecary's profit maximizing price?

Q3 | Maximum Profit

What is the Apothecary's maximum profit?